

ping apart the conventional soldier's gear, armour and clothing, and then working from scratch. The primary constraint here is weight. Studies have shown that in battle, soldiers tend to throw away gear that is too heavy. The system is designed so that the components are extremely portable and light, yet rugged enough to withstand battle conditions. Finally, to put together the whole system, all future combat systems are planned in such a way that they are built over several upgradeable subsystems—the integrated helmet subsystem, the computer and communication subsystem, the weapons subsystem, the protective clothing, and the software that powers the hardware. The infantry will see integrated weapons systems, such as satellite linked radio, rifle mounted thermal sensors and video cameras, and GPS systems, all of which will give them the edge and tell them where they are and more importantly, where the enemy is.

Sarge... I see unfriendlies in my HUD...

Today's warriors jump into combat situations armed with increasingly sophisticated weapons, but with information that is at least a few hours old. Things move amazingly fast in battle, and soldiers are put at risk if they are just dropped into a hot zone with vague information. The Integrated Helmet Assembly Subsystem

(IHAS) is straight out of a sci-fi movie—in the battle zone of the future, the very visible helmet and helmet-mounted gear, is going to be the soldier's networking and communication device. The prototype feeds up-to-date information regarding a combat zone into a HUD (Heads-Up Display), straight off a satellite. Initial systems that will rapidly update tactical information on a map-like layout, and detail the positions of squad mates are on the drawing board. The optical systems come with daylight cameras that will transmit everything a soldier sees, in real time. Additionally, there are lightweight thermal sights, night vision, as well as ten digit GPS and grid coordinate displays. Soldiers will no longer have to carry multiple sights—thermal vision, night vision and so on. Researchers claim that the image that the soldiers will see in the full colour SVGA HUD micro-display should be equivalent to two 17-inch displays right in front of their eyes!

But there is more to come for the HUD, especially for the Objective Force Warrior. Chemical and biological sensors as well as 3D imaging systems, will allow a soldier to view his surroundings from different angles, such as from a squad mate's perspective. The initial systems will allow them to view images and video straight from the weapon mounted sighting cameras. With the HUD integrated with the warrior's rifle sights and firing system, the soldier could accurately fire his assault rifle or spot an inaccessible target and call for indirect fire.

War zones are not the place for delicate instruments, thus the computer and communications subsystem is designed to take a beating and yet continue working. Interestingly, the initial prototypes were developed using a mobile Pentium III class processor, boasted of USB connectivity and ran on a specially developed version of Windows 2000. The computer comes with a solid-state Flash drive with 800 MB of storage, which stores maps and overlays, graphics, battle field information, orders and tactical aids. There will also be real time intelligence getting piped into the flash drive that will be rapidly updated.

But this is just half of the computer and communications subsystem. The other half is what soldiers call the 'battle rattle'—the wireless LAN that networks all the Land Warriors. Everything that a soldier sees, is transmitted from the display mounted on his helmet to all other team

Sorry Sarge, I'm a techno-ignoramus...

Today's trained warriors are as much at ease with a computer, or taking down a network using hacking tools, as they are with stripping apart an assault rifle and putting it back together. Militaries around the world are dealing with a new exercise in warfare—and the turf is the Internet. No army can afford to have soldiers that can't bring down any enemy's communication networks, or repair a network outage whilst under fire. Cyber warfare involves anticipating, averting, and responding to cyber terrorism attacks that try to yank down world-wide communications networks.

Future programs will look into bringing down all possible infrastructure that has been hooked to the Internet. The US military training schools have been actively training cadets in cyber warfare, and so have other countries, such as China. Most of us believe that redundancy is built into the Internet, and that with multiple nodes, traffic can be re-routed to ensure that the network is always up. But, recent attacks on key nodes have shown that concerted attacks can succeed. The battlegrounds of the 21st century will include the Internet.

members, as well as the control centres, via a satellite hook-up. At the heart of the communication box is an Intel StrongARM RISC processor that runs Windows—in this case, Windows CE. The hook-up to the LAN is achieved through a wireless LAN card. Under battle conditions, messages can be relayed at 1.3 Mbps up to a range of 1.3 Km—research is currently on to raise the limit to 2 Mbps at 5 Km. The network traffic is encrypted using military grade encryption and authentication codes.

To find out just where the soldiers are located, the navigations system runs a military grade GPS receiver that tracks the precise geographical coordinates of all the warriors in action. Backing this up is an inertial navigation module that kicks in where the GPS satellite signals don't penetrate—such as deep inside a jungle or in city streets between tall buildings.

A big hiccup are power supplies, which have to be able to withstand combat conditions. Currently, cells have limited endurance and tend to be heavy. The Land Warrior system draws juice from



This is how the land warrior is going to be outfitted